

Practical 4

Problem Definition

Add dynamic UI components such as collapsible FAQ, modal popup, image/content slider, notification banner, hamburger menu, and light/dark theme switcher using JavaScript.

KEY QUESTIONS

1. How are DOM elements selected and modified?

DOM elements are selected using JavaScript methods like `querySelector()` and `getElementById()`. Their content, style, and visibility are modified using JavaScript.

2. Are event listeners attached correctly?

Yes. Event listeners are attached to buttons and other UI elements. They perform actions when the user clicks or interacts with them.

3. Is `localStorage` used for remembering theme choice?

Yes. `localStorage` is used to save the user's light or dark theme choice, so the selected theme remains after refreshing or reopening the page.

4. Does interactivity improve usability without breaking accessibility?

Yes. Interactive components make the website easier and more user-friendly. Proper buttons, labels, and simple controls are used to keep the features accessible.

B) code

about.js

```
alert("Welcome to Student Hub Portal!");

console.log("About Page Loaded Successfully");

document.querySelector("h1").innerHTML = "Welcome to Student Hub Portal";

let features = document.querySelectorAll("ul li");

console.log("Total Features : " + features.length);

document.querySelector("h1").onclick = function () {
```

```
        this.style.color = "red";
    };

    document.querySelector("img").onmouseover = function () {

        this.style.border = "5px solid blue";

    };

    document.querySelector("img").onmouseout = function () {

        this.style.border = "2px solid black";

    };
};
```

assignment.js

```
alert("Welcome to Assignment Page");
console.log("Assignment Page Loaded");
let submitBtn = document.querySelector('input[type="submit"]');
let resetBtn = document.querySelector('input[type="reset"]');
let fileInput = document.querySelector('input[type="file"]');
submitBtn.onclick = function (event) {
    event.preventDefault();
    if (fileInput.value == "") {
        alert("Please select an assignment file.");
    } else {
        alert("Assignment Submitted Successfully!");
    }
};
resetBtn.onclick = function () {
    alert("Form Reset Successfully!");
};
fileInput.onChange = function () {
    alert("File Selected Successfully!");
};
```

attendance.js

```
alert("Welcome to Attendance Page");
console.log("Attendance Page Loaded");
```

```
let submitBtn = document.querySelector('input[type="submit"]');
let resetBtn = document.querySelector('input[type="reset"]');
let subjectInput = document.querySelector('input[type="text"]');
let rows = document.querySelectorAll("table tr");
submitBtn.onclick = function(event) {
    event.preventDefault();
    let subject = subjectInput.value.toLowerCase();
    let found = false;
    for (let i = 1; i < rows.length; i++) {
        let subjectName = rows[i].cells[1].innerText.toLowerCase();
        if (subjectName.includes(subject)) {
            rows[i].style.display = "";
            found = true;
        } else {
            rows[i].style.display = "none";
        }
    }
    if (!found) {
        alert("Subject not found!");
    }
};
resetBtn.onclick = function() {
    subjectInput.value = "";
    for (let i = 1; i < rows.length; i++) {
        rows[i].style.display = "";
    }
    alert("Search Reset Successfully!");
};
```

[contact.js](#)

```
function validateForm() {
    let name = document.getElementById("name").value;
    let email = document.getElementById("email").value;
    let phone = document.getElementById("phone").value;
    let message = document.getElementById("message").value;
    if (name == "") {
        alert("Please enter your name");
        return false;
    }
    if (email == "") {
```

```
    alert("Please enter your email");
    return false;
  }
  if (phone == "") {
    alert("Please enter your phone number");
    return false;
  }
  if (message == "") {
    alert("Please enter your message");
    return false;
  }
  alert("Form submitted successfully!");
  return false;
}
```

course.js

```
function searchCourse() {
let search = document.getElementById("searchCourse").value.toLowerCase();
    let rows =
document.getElementById("courseTable").getElementsByName("tr");
    for (let i = 1; i < rows.length; i++) {
        let courseID =
rows[i].getElementsByName("td")[0].innerText.toLowerCase();
        let courseName =
rows[i].getElementsByName("td")[1].innerText.toLowerCase();
        if (courseID.includes(search) || courseName.includes(search)) {
            rows[i].style.display = "";
        }
        else {
            rows[i].style.display = "none";
        }
    }
}
```

login.js

```
function login() {
    let email = document.getElementById("email").value;
    let password = document.getElementById("password").value;
    if (email == "" || password == "") {
        alert("Please enter email and password");
    }
}
```

```
}  
    else if (email == "student@gmail.com" && password == "12345") {  
        alert("Login Successful");  
        window.location.href = "home.html";  
    }  
    else {  
        alert("Wrong email or password");  
    }  
}
```

profile.js

```
function submitProfile() {  
    let name = document.getElementById("name").value;  
    let enrollment = document.getElementById("enrollment").value;  
    let email = document.getElementById("email").value;  
    let mobile = document.getElementById("mobile").value;  
    if (name == "") {  
        alert("Please enter your full name");  
        return false;  
    }  
    if (enrollment == "") {  
        alert("Please enter your enrollment number");  
        return false;  
    }  
    if (email == "") {  
        alert("Please enter your email");  
        return false;  
    }  
    if (mobile == "") {  
        alert("Please enter your mobile number");  
        return false;  
    }  
    alert("Profile Submitted Successfully!");  
    return false;  
}
```

result.js

```
function checkResult() {  
    let enrollment = document.getElementById("enrollment").value;  
    if (enrollment == "") {
```

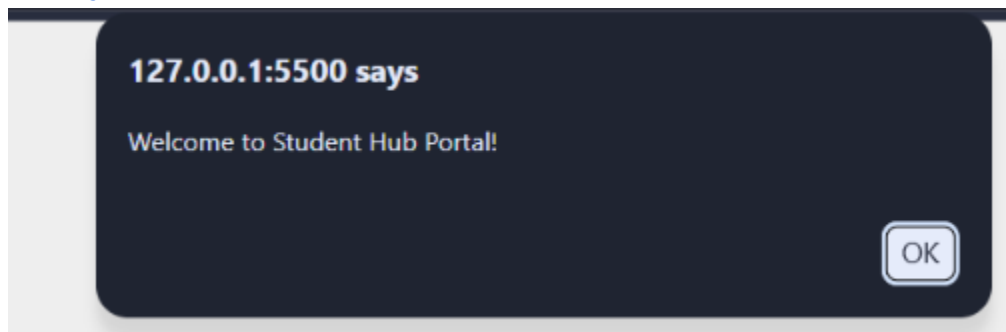
```
    alert("Please enter enrollment number");  
  }  
  else {  
    alert("Result found successfully!");  
  }  
}
```

[studentfeedback.js](#)

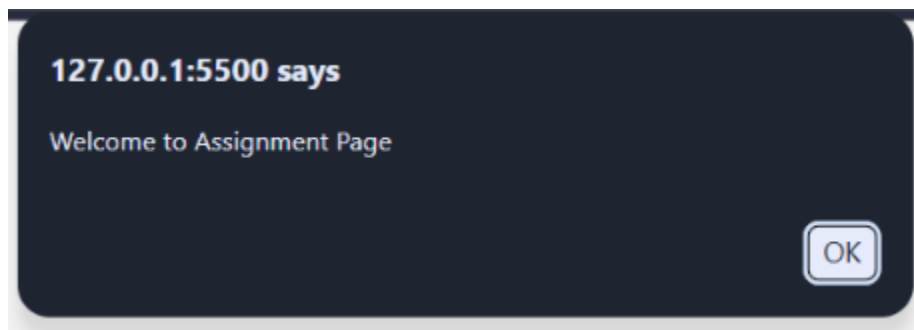
```
function submitFeedback() {  
    alert("Feedback submitted successfully!");  
}
```

c)output

[about.js](#)



[assignment.js](#)



127.0.0.1:5500 says
File Selected Successfully!

OK

Assignment No	Assignment Name	Assignment Description	Assignment Submission Date
1	Assignment 1		2024-06-30
2	Assignment 2	This is Assignment 2 description	2024-07-15
3	Assignment 3	This is Assignment 3 description	2024-08-01

Upload Assignment

Choose File

No file chosen

Submit Assignment

Reset

About

Assignment

Attendance

Contact

Course

Login

Profile

Result

Students Feedback

Dashboard

attendance.js

127.0.0.1:5500 says
Welcome to Attendance Page

OK

student-hub attendance portal

date	subject	total lectures	lectures attended	lecture absent	attendance percentage
2024-06-02	java	15	12	3	80%

search subject:

java

Submit

Reset

about

assignment

attendance

contact

course

login

profile

result

studentsfeedback

dashboard

contact.js

127.0.0.1:5500 says
Form submitted successfully!

OK

Email:

nency1709@gmail.com

Phone:

9913872030

Message:

no

Submit

Reset

[course.js](#)

LOGO

[About](#) [Assignment](#) [Attendance](#) [Contact](#) [Course](#) [Login](#) [Profile](#) [Result](#) [Students Feedback](#) [Dashboard](#)

Available Courses

java

Course ID	Course Name	Duration	Faculty
CEUC:203	Java Programming	6 Months	Mr. Deven Gol

Student Hub Portal
25dce023@charusat.edu.in

[login.js](#)

127.0.0.1:5500 says
Wrong email or password

OK

Login

☐ Remember Me

SubmitReset

Create an Account

Forgot Password

[profile.js](#)

127.0.0.1:5500 says
Profile Submitted Successfully!

OK

9913872030

Date of Birth
25 - 08 - 2026

Gender
☐ Male ☒ Female ☐ Other

Course
Computer Science

Semester
3rd semester

Address
no

Submit

[result.js](#)

127.0.0.1:5500 says
Result found successfully!

OK

Semester:

Sem 1

Submit

Reset

Subject	Marks Obtained	Total Marks	Result
Mathematics	90	100	Pass
Java	80	100	Pass
DSA	85	100	Pass
CN	85	100	Pass
DSA	85	100	Pass

[Dashboard](#)

[feedback.js](#)

127.0.0.1:5500 says
Feedback submitted successfully!

OK

Faculty Name:

khushi mam

Rating:

Excellent

Feedback:

no

Submit

Reset

D) github repository link

[Studenthub/js at main · nency1709/Studenthub](https://github.com/nency1709/Studenthub)

Problems Faced

- Understanding how to use JavaScript with HTML elements.
- Selecting and modifying DOM elements correctly.
- Handling click events for buttons and menus.
- Implementing the image/content slider smoothly.
- Managing the open/close behavior of FAQ, modal, and hamburger menu.
- Storing the light/dark theme preference using **localStorage**.
- Making the dynamic components user-friendly and accessible.

Learning Outcomes

- Learned how to manipulate HTML elements using JavaScript DOM.
- Learned how to use event listeners for user interactions.
- Learned to create interactive FAQ, modal popup, slider, and hamburger menu.
- Learned how to implement light/dark mode using JavaScript.
- Learned how to use localStorage to remember user preferences.
- Improved understanding of creating dynamic and user-friendly web pages.